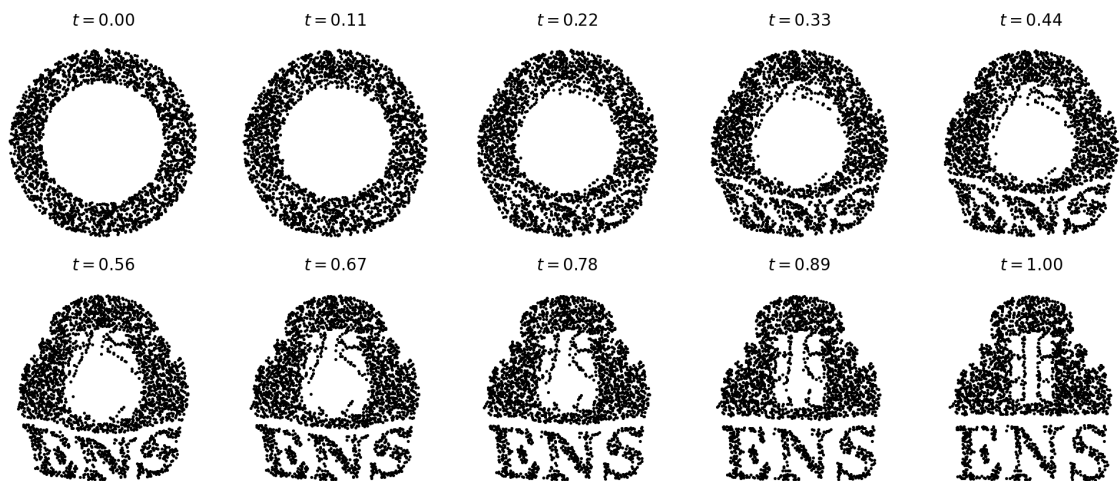


ÉCOLE NORMALE SUPÉRIEURE

Mathematics Thesis

Wasserstein Barycenters



François Sarcos

Supervised by Julie Delon

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1. Preliminaries

1.1 Introduction

Optimal transport theory has made it possible to compare probability measures precisely, which has paved the way for numerous applications, first military and economic, then in data science, image processing and machine learning, to name but a few. Wasserstein barycenters generalize the well-known notion of the Euclidean barycenter of a set of points to that of a set of measures. In this thesis, we study the barycenter for the W_2 distance of p probability measures on $\mathbb{R}^d$, relying mainly on the work of Martial Agueh and Guillaume Carlier [1]. First, we prove existence and uniqueness results for such a barycenter using a dual problem. Next, we present the link between our barycenter problem and a generalization of the transport problem to an arbitrary number of measures (multi-marginal formulation). Finally, we make the solutions explicit in particular cases,

such as that of two measures, where the barycenter gives the shape of the geodesics in the space under study, or that of p Gaussians, in which the barycenter is itself Gaussian, and we present how to compute these barycenters numerically.

1.2 Notations

We introduce the space Y :

$$Y := \left\{ f \in C(\mathbb{R}^d) : \frac{f}{1 + |\cdot|^2} \text{ is bounded} \right\}$$

endowed with the norm:

$$\|f\|_Y := \sup_{x \in \mathbb{R}^d} \frac{|f(x)|}{1 + |x|^2}$$

Let X be the closed subspace of Y :

$$X := \left\{ f \in C(\mathbb{R}^d) : \lim_{|x| \rightarrow \infty} \frac{f(x)}{1 + |x|^2} = 0 \right\}$$

We denote by $\mathcal{M}(\mathbb{R}^d)$ the space of finite signed measures on $\mathbb{R}^d$, which we identify with the dual of the space $C_0(\mathbb{R}^d)$ of functions vanishing at infinity.

We denote by $\mathcal{M}_+^1(\mathbb{R}^d)$ the set of probability measures. We will identify the dual of X with:

$$X' := \{ \mu \in \mathcal{M}(\mathbb{R}^d) : (1 + |x|^2) \mu \in \mathcal{M}(\mathbb{R}^d) \}$$

For two probability measures ν and μ with finite second moments, we denote by $W_2(\mu, \nu)$ the Wasserstein distance between μ and ν , which is by definition the cost associated with the optimal transport between μ and ν for the Kantorovich problem. It is indeed a distance in the sense of metric spaces. More precisely, denoting by $\Pi(\mu, \nu)$ the set of transport plans from μ to ν , we have:

$$W_2^2(\mu, \nu) := \inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 d\gamma(x, y)$$

We fix an integer $p \geq 2$, nonnegative scalars $\lambda_1, \dots, \lambda_p$ such that $\sum_{i=1}^p \lambda_i = 1$ and probability measures $\nu_1, \dots, \nu_p$ with finite second moments, that is $\nu_i \in X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$. We want to solve the problem:

$$(P) \quad \inf_{\nu \in X' \cap \mathcal{M}_+^1(\mathbb{R}^d)} J(\nu) := \sum_{i=1}^p \frac{\lambda_i}{2} W_2^2(\nu_i, \nu) \quad (1.1)$$

1.3 Some useful results

We present here some classical optimal transport results that will be very useful, some of which are taken for granted here, but whose proofs can be found in Filippo Santambrogio's book [2].

Definition 1.1

When ϕ is a function $\mathbb{R}^d \rightarrow \mathbb{R}$, we denote by $\partial\phi$ the set:

$$\partial\phi := \{(x, y) : x \in \mathbb{R}^d, y \in \partial\phi(x)\}$$

Remark 1.2. This set is, in a sense, the graph of the subdifferential of ϕ .

Theorem 1.3: Characterization of the solution of the transport problem

Let $\mu, \nu \in X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$ and $\gamma \in \Pi(\mu, \nu)$.

Then γ is an optimal transport plan from μ to ν if and only if there exists a convex l.s.c. function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $\text{supp}(\gamma) \subset \partial\phi$. In this case, ϕ attains the infimum of the problem:

$$\inf \left\{ \int_{\mathbb{R}^d} \phi(x) d\mu(x) + \int_{\mathbb{R}^d} \phi^*(y) d\nu(y) : \phi \text{ convex l.s.c.} \right\} \quad (1.2)$$

where ϕ^* denotes the Legendre transform of ϕ . If moreover μ is absolutely continuous with respect to the Lebesgue measure, then γ is unique and given by:

$$\gamma = (\text{id}, \nabla\phi) \# \mu \quad (1.3)$$

Remark 1.4. The assumption of absolute continuity with respect to the Lebesgue measure ensures that ϕ is differentiable μ -almost everywhere.

Definition 1.5

When μ is absolutely continuous with respect to the Lebesgue measure and in $X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$, we call **Brenier map** from μ to ν the map $\nabla\phi$ provided by the previous theorem.

We know how to perform the transport in the reverse direction thanks to the following theorem:

Theorem 1.6

When μ and ν are measures that are absolutely continuous with respect to the Lebesgue measure and in $X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$, denoting by ϕ the Brenier map from μ to ν , we have:

The Legendre transform ϕ^* is differentiable ν -almost everywhere, and it is the Brenier map from ν to μ . Moreover, μ -a.e.: $\nabla\phi^* \circ \nabla\phi = \text{id}$.

Finally, we can make the Brenier map explicit in the case of two Gaussians:

Theorem 1.7

Let $\Sigma_1, \Sigma_2 \in \mathcal{S}_n^{++}(\mathbb{R})$. When $\mu \sim \mathcal{N}(0, \Sigma_1)$ and $\nu \sim \mathcal{N}(0, \Sigma_2)$, the Brenier map T from μ to ν is given by:

$$T = \Sigma_2^{1/2} \left(\Sigma_2^{1/2} \Sigma_1 \Sigma_2^{1/2} \right)^{-1/2} \Sigma_2^{1/2}$$

Proof. By [theorem 1.3](#), it suffices to check that T can be written $\nabla\phi$ where ϕ is a convex l.s.c. function and that $T \# \mu = \nu$. For the first point, we take $\phi : x \mapsto 1/2 \langle Tx, x \rangle$. For the second point, performing the bijective change of variables $y = T^{-1}x$, whose Jacobian is T^{-1} , with $\det T^{-1} = |\Sigma_2|^{-1/2} |\Sigma_1|^{1/2}$,

we have for measurable A :

$$\begin{aligned}
T\sharp\mu(A) &= \mu(T^{-1}(A)) \\
&= \frac{1}{\sqrt{2\pi}^d |\Sigma_1|^{1/2}} \int_{T^{-1}(A)} e^{-\frac{1}{2}\langle \Sigma_1^{-1}y, y \rangle} d\lambda(y) \\
&= \frac{1}{\sqrt{2\pi}^d |\Sigma_2|^{1/2}} \int_A e^{-\frac{1}{2}\langle \Sigma_1^{-1}T^{-1}x, T^{-1}x \rangle} \det T^{-1} d\lambda(x)
\end{aligned}$$

But using that Σ_1 and Σ_2 are symmetric and invertible, we have:

$$\langle \Sigma_1^{-1}T^{-1}x, T^{-1}x \rangle = \langle \Sigma_2^{-1}x, x \rangle$$

Hence:

$$T\sharp\mu(A) = \frac{1}{\sqrt{2\pi}^d |\Sigma_2|^{1/2}} \int_A e^{-\frac{1}{2}\langle \Sigma_2^{-1}x, x \rangle} d\lambda(x) = \nu(A)$$

■

2. Study of the primal problem (P) and its dual

In this part, we focus on problem (P) . Note that solving problem (P) amounts to finding (provided a solution exists) a barycenter ν of the set of measures $\nu_1, \dots, \nu_p$ associated with the weights $\lambda_1, \dots, \lambda_p$. Indeed, the factor $1/2$ in (1.1) does not change the argmin of (P) , and W_2 is a distance on the space $X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$ of probability measures with finite second moment. Moreover, given the cost function used in the integral defining W_2 , it is clear that this barycenter is the analogue for measures of the usual Euclidean barycenter of points $x_1, \dots, x_p$ of $\mathbb{R}^d$ associated with the weights $\lambda_1, \dots, \lambda_p$ and defined as:

$$x^* := \operatorname{argmin}_{x \in \mathbb{R}^d} \sum_{i=1}^p \lambda_i \|x - x_i\|^2$$

2.1 Duality

We now introduce the dual problem, which we will use to prove the uniqueness of the solution of (P) , as well as the functional S_λ .

Definition 2.1

For $f \in Y$, we define:

$$S_\lambda f(x) := \inf_{y \in \mathbb{R}^d} \frac{\lambda}{2} |x - y|^2 - f(y) \quad (2.1)$$

Proposition 2.2

The dual problem of (P) is the problem (P_0^*) defined by:

$$(P_0^*) \quad \sup_{f_i \in X, \sum_{i=1}^p f_i = 0} F(f_1, \dots, f_p) := \sum_{i=1}^p \int_{\mathbb{R}^d} S_{\lambda_i} f_i d\nu_i \quad (2.2)$$

Proof. Theorem 2.5 will justify this duality. ■

However, the supremum of (P^*) may not be attained, hence the introduction of the relaxed problem (P^*) :

Definition 2.3

We consider the problem:

$$(P^*) \quad \sup_{f_i \in Y, \sum_{i=1}^p f_i = 0} F(f_1, \dots, f_p) \quad (2.3)$$

Proposition 2.4: Weak duality

We have the inequalities:

$$\inf(P) \geq \sup(P^*) \geq \sup(P_0^*) \quad (2.4)$$

Proof. By definition of S_{λ_i} we have, for all x_i, y in $\mathbb{R}^d$ and f_i in Y :

$$S_{\lambda_i} f(x) + f_i(y) \leq \frac{\lambda_i}{2} |x_i - y|^2$$

Integrating with respect to γ_i and summing over i , we get:

$$\sum_{i=1}^p \int_{\mathbb{R}^d} S_{\lambda_i} f_i d\gamma_i \leq \sum_{i=1}^p \frac{\lambda_i}{2} \int_{\mathbb{R}^d} |x_i - y|^2 d\gamma_i(x_i, y)$$

Taking the sup and the inf, we obtain the first inequality. The second inequality is due to the fact that X is included in Y . ■

Theorem 2.5: Strong duality

We have the equalities:

$$\inf(P) = \sup(P^*) = \sup(P_0^*) \quad (2.5)$$

Before proving the theorem, let us introduce the functional H_i and its Legendre transform:

Definition 2.6

For $f \in X$ and $\nu \in X'$ we set:

$$H_i(f) := - \int_{\mathbb{R}^d} S_{\lambda_i} f d\nu_i \quad (2.6)$$

$$H_i^*(\nu) := \sup_{f \in X} \int_{\mathbb{R}^d} f d\nu - H_i(f) = \sup_{f \in X} \int_{\mathbb{R}^d} f d\nu + \int_{\mathbb{R}^d} S_{\lambda_i} f d\nu_i \quad (2.7)$$

$$H_i^{**}(f) := \sup_{\nu \in X'} \int_{\mathbb{R}^d} f d\nu - H_i^*(\nu) \quad (2.8)$$

Lemma 2.7

Let $\nu \in X'$

$$H_i^*(\nu) = \begin{cases} \frac{\lambda_i}{2} W_2^2(\nu_i, \nu) & \text{if } \nu \in X' \cap \mathcal{M}_+^1(\mathbb{R}^d) \\ +\infty & \text{otherwise} \end{cases} \quad (2.9)$$

Proof. Suppose that $\nu \notin X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$. Since $\nu \in X'$, either ν is not positive, or it does not have mass 1.

In the case where ν is not positive, there exists a nonpositive $f \in X$ such that $\int_{\mathbb{R}^d} f d\nu > 0$. But since f is nonpositive, given the definition of S_{λ_i} , for all $t \geq 0$, $S_{\lambda_i}(tf) \geq 0$. Hence $H_i^*(f) \geq \int_{\mathbb{R}^d} tf d\nu$. We let t tend to infinity and, since we took f with strictly positive integral, we have: $H_i^*(f) \geq +\infty$. In the case where ν is positive but does not have mass 1, we may assume without loss of generality that $\nu(\mathbb{R}^d) < 1$. Fix $t \geq 0$ and define the function $f(x) = -t$, which is indeed in X . We have:

$$H_i^*(f) \geq -t \left(\int_{\mathbb{R}^d} d\nu + \int_{\mathbb{R}^d} d\nu_i \right) = t(1 - \nu(\mathbb{R}^d)).$$

Then we let t tend to $+\infty$.

Now suppose that $\nu \in X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$. By Kantorovich duality, we have:

$$\begin{aligned} \frac{\lambda_i}{2} W_2^2(\nu, \nu_i) &= \sup \left\{ \int_{\mathbb{R}^d} f d\nu + \int_{\mathbb{R}^d} g d\nu_i \mid f, g \in C_b, f(x) + g(y) \leq \frac{\lambda_i}{2} |x - y|^2 \right\} \\ &= \sup_{f \in X} \left\{ \int_{\mathbb{R}^d} f d\nu + \int_{\mathbb{R}^d} S_{\lambda_i} f d\nu_i \right\} = H_i^*(\nu) \end{aligned}$$

■

We can now prove strong duality:

Proof. By lemma 2.7:

$$\inf(P) = \inf_{\nu \in X'} \sum_{i=1}^p H_i^*(\nu) = - \left(\sum_{i=1}^p H_i^* \right)^* \quad (0)$$

We define, for $f \in X$:

$$\mathcal{H}(f) := \inf_{\substack{f_i \in X \\ \sum_{i=1}^p f_i = f}} \sum_{i=1}^p H_i(f_i) \quad (2.10)$$

Then:

$$\sup(P_0^*) = -\mathcal{H}(0)$$

And by a simple rewriting, we have:

$$\mathcal{H}^* = \sum_{i=1}^p H_i^*$$

Weak duality therefore tells us that:

$$-\mathcal{H}^{**}(0) = \inf(P) \geq \sup(P_0^*) = -\mathcal{H}(0)$$

It therefore suffices to prove that $\mathcal{H}(0) = \mathcal{H}^{**}(0)$ to complete the proof. $\mathcal{H}$ is convex because the H_i are, as suprema of affine functions (in the variable f_i), and the inequality expressing the convexity of the H_i carries over to the inf. Since $\mathcal{H}$ is convex, it remains to show that $\mathcal{H}$ is l.s.c. and does not take the value $-\infty$. For this we use the fact that a convex function on a normed vector space that is bounded above in a neighborhood of 0 and admits a continuous affine minorant is continuous.

- **Affine minorant:** Let $f \in X$. Taking $y = 0$ in (2.1), we get:

$$H_i(f_i) \geq f_i(0) - \frac{\lambda_i}{2} \int_{\mathbb{R}} |x|^2 dv_i(x)$$

where the integral is finite because we took $v_i \in X'$. Hence:

$$\mathcal{H}(f) \geq f(0) - \sum_{i=1}^p \frac{\lambda_i}{2} \int_{\mathbb{R}} |x|^2 dv_i(x)$$

This incidentally gives us that $\mathcal{H} > -\infty$.

- **Upper bound in a neighborhood of 0:** Let $f \in X$ be such that $\|f\|_Y \leq \frac{p}{4} \min_i(\lambda_i)$. We have:

$$\begin{aligned} H(f) &\leq \sum_{i=1}^p H_i\left(\frac{f}{p}\right) \leq \sum_{i=1}^p \int_{\mathbb{R}^d} \sup_{y \in \mathbb{R}^d} \left\{ \frac{\lambda_i}{4} (1 + |y|^2) - \frac{\lambda_i}{2} |x - y|^2 \right\} dv_i(x) \\ &= \sum_{i=1}^p \int_{\mathbb{R}^d} \left(\frac{\lambda_i}{4} + \frac{\lambda_i}{2} |x|^2 \right) dv_i(x) \\ &= \frac{1}{4} + \sum_{i=1}^p \frac{\lambda_i}{2} \int_{\mathbb{R}^d} |x|^2 dv_i(x). \end{aligned}$$

In the first inequality, we use the definition of $\mathcal{H}$; in the second, we use the definition of S_λ and the assumption on the norm of f . In the last one, we use the fact that the v_i have mass 1. Finally, v_i is in X' , so the upper bound is finite.

■

2.2 Existence of solutions

Theorem 2.8: Existence of solutions

The infimum of (P) and the supremum of (P^*) are attained.

2.2.1 Existence for (P)

For (P) the proof is more elementary and relies on Prokhorov's theorem, which we recall in the case of interest to us.

Theorem 2.9: Prokhorov

Let (v^n) be a sequence with values in $\mathcal{M}_1^+(\mathbb{R}^d)$. There exist $\nu \in \mathcal{M}_1^+(\mathbb{R}^d)$ and a subsequence of (v^n) converging narrowly to ν if and only if (v^n) is tight.

Remark 2.10. Recall that a set Γ is said to be “tight” if $\forall \varepsilon > 0, \exists K$ a compact set such that $\sup_{\mu \in \Gamma} \mu(K^c) \leq \varepsilon$

Remark 2.11. Compared with the general case, we use here the fact that $\mathcal{M}_1^+(\mathbb{R}^d)$ is narrowly closed and is a bounded subset of a space in duality with a metrizable space $(C_0(\mathbb{R}^d))$.

We can now prove the existence of a minimizer for (P) .

Proof. Let (v^n) be a minimizing sequence for (P) . The sequence $(J(v^n))_n$ is therefore bounded above. Hence, for all i , $(W_2^2(v_i, v^n))_n$ is bounded above. Now, we know that there exists an optimal transport plan $\gamma_{n,i}$ from v_i to v^n , so:

$$\begin{aligned} W_2^2(v_i, v^n) &= \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 d\gamma(x, y) \\ &\geq \int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{1}{2} |x|^2 d\gamma(x, y) - \int_{\mathbb{R}^d \times \mathbb{R}^d} |y|^2 d\gamma(x, y) \\ &= \frac{1}{2} \int_{\mathbb{R}^d} |x|^2 dv^n(x) - \int_{\mathbb{R}^d} |y|^2 dv_i(y). \end{aligned}$$

This shows that $\left(\int_{\mathbb{R}^d} |x|^2 dv^n(x) \right)_n$ is bounded. Markov’s inequality (applied to “the r.v. $|X|^2$ ”) gives, for $R > 0$:

$$v^n(|x| > R) \leq \frac{1}{R^2} \int_{\mathbb{R}^d} |x|^2 dv^n(x)$$

By Prokhorov’s theorem, up to extraction, we may assume that (v^n) converges narrowly to v . Let f_R be the bounded function $f_R(x) := |x|^2 \mathbf{1}_{|x|^2 \leq R}$. This function is not continuous, but we can modify f slightly at the boundary to make it continuous. This does not change the reasoning that follows. We have:

$$\int_{\mathbb{R}^d} f_R dv^n \leq \int_{\mathbb{R}^d} |x|^2 dv^n(x)$$

But by narrow convergence:

$$\int_{\mathbb{R}^d} f_R dv^n \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}^d} f_R dv$$

By monotone convergence:

$$\int_{\mathbb{R}^d} f_R dv \xrightarrow{R \rightarrow \infty} \int_{\mathbb{R}^d} |x|^2 dv(x)$$

Hence the boundedness of the sequence of second moments ensures that $v \in X'$. Moreover, since narrow convergence is stronger than weak- * convergence, (v_n) converges weak- * to v . To conclude, it remains to show that J is l.s.c. for weak- * convergence. But [lemma 2.7](#) concludes, since H_i^* is a Legendre transform and is therefore weak- * l.s.c. ■

2.2.2 Existence for (P^*)

Let us now show the existence of a maximizer for (P^*) . We know that $\sup(P^*) = \sup(P_0^*)$. So if we take a maximizing sequence for problem (P_0^*) , it is also a maximizing sequence for (P^*) . This observation is very important: it tells us that we are free to impose a strong condition on our f_i^n , namely that of belonging to X , which is crucial for the reasoning that follows. The procedure is as follows:

- Notice that we can impose on the f_i^n constraints that are even stronger than being in X (lemma 2.13).
- Show that the sequence (f_i^n) is bounded in Y .
- Show that we can extract a subsequence converging pointwise to some f_i in Y .
- Show that the function being maximized is u.s.c.

To lighten notation, we allow ourselves to omit the indices i or n when this causes no ambiguity.

Lemma 2.12

S_λ maps X into itself.

Proof. We start by noting that, for all $x \in \mathbb{R}^d$,

$$S_\lambda f(x) = \inf_{y \in \mathbb{R}^d} \left\{ \frac{\lambda}{2} |x - y|^2 - f(y) \right\} \leq -f(x),$$

by choosing $y = x$. We deduce that

$$\limsup_{|x| \rightarrow \infty} \frac{S_\lambda f(x)}{|x|^2} \leq 0.$$

Moreover, since $f \in X$, for all $\varepsilon > 0$, there exists a constant $C \in \mathbb{R}$ such that

$$\forall y \in \mathbb{R}^d, \quad f(y) \leq \varepsilon |y|^2 + C.$$

Thus,

$$\begin{aligned} S_\lambda f(x) &= \inf_{y \in \mathbb{R}^d} \left\{ \frac{\lambda}{2} |x - y|^2 - f(y) \right\} \\ &\geq \inf_{y \in \mathbb{R}^d} \left\{ \frac{\lambda}{2} |x - y|^2 - \varepsilon |y|^2 - C \right\}. \end{aligned}$$

If $\varepsilon < \frac{\lambda}{2}$, the function

$$y \mapsto \frac{\lambda}{2} |x - y|^2 - \varepsilon |y|^2$$

is a parabola, whose minimum can be computed explicitly. It is attained at a point depending affinely on x , and we deduce the existence of constants $D, E \in \mathbb{R}$ such that

$$S_\lambda f(x) \geq -D|x|^2 + E.$$

We then obtain

$$\liminf_{|x| \rightarrow \infty} \frac{S_\lambda f(x)}{|x|^2} \geq 0.$$

■

Lemma 2.13

There exist $f_i^n \in X$ such that $(f_1^n, \dots, f_p^n)$ is a maximizing sequence of (P^*) , and $g_i^n \in X$ satisfying the following conditions:

- (i) $f_i^n(0) = 0 \quad \forall i$
- (ii) $f_i^n = S_{\lambda_i} g_i^n \quad \forall i \leq p-1$
- (iii) $g_i^n = S_{\lambda_i} f_i^n \quad \forall i$

Proof. There exists a sequence $(h_1^n, \dots, h_p^n)$ of elements of X , maximizing for (P_0^*) , which is also maximizing for (P^*) , as noted above. Let us set, for all i, n :

$$g_i^n := S_{\lambda_i} h_i^n, \quad \tilde{f}_i^n := S_{\lambda_i} g_i^n$$

We check that:

- $h_i \leq \tilde{f}_i$: Indeed, for fixed x :

$$\begin{aligned} h(x) \leq \tilde{f}(x) &\Leftrightarrow \forall y, \quad h(x) \leq \frac{\lambda}{2}|x-y|^2 - g(y) \\ &\Leftrightarrow \forall y, \quad g(y) \leq \frac{\lambda}{2}|x-y|^2 - h(x) \end{aligned}$$

- $S_\lambda \tilde{f} = S_\lambda h$:

On the one hand, $h \leq \tilde{f}$, so $S_\lambda \tilde{f} \leq S_\lambda h$. On the other hand: $S_\lambda \tilde{f} = (S_\lambda S_\lambda) S_\lambda h \geq S_\lambda h$ (where the inequality follows from the fact that applying S_λ twice increases the value. This is, more or less, what we check in the claim: $h_i \leq \tilde{f}_i$.)

We define:

$$(f_1^n, \dots, f_{p-1}^n) := (\tilde{f}_1^n, \dots, \tilde{f}_{p-1}^n) \quad f_p^n := - \sum_{i=1}^{p-1} \tilde{f}_i^n$$

$g_i^n \in X$: lemma 2.12; $f_i^n \in X$: lemma 2.12; $\sum_{i=1}^p f_i^n = 0$: by construction; **(i)**: we can subtract constants from the f_i , as long as their sum remains zero, without changing the functional F ; **(ii)**: true by definition; **(iii)**: this follows from $S_\lambda \tilde{f} = S_\lambda h$; $F(f_1, \dots, f_p) \geq F(h_1, \dots, h_p)$: for $i \leq p-1$, $h_i \leq f_i$ and $f_p \leq \sum_{i=1}^{p-1} h_i = h_p$. Hence, for all i , $S_{\lambda_i} f_i \geq S_{\lambda_i} h_i$. We conclude by monotonicity of the integral. ■

Let us prove the existence of a maximizer:

Proof. Let f_i^n, g_i^n be as in lemma 2.13. Expanding the scalar product, we have:

$$\frac{\lambda_i}{2}|y|^2 - f_i(y) = \sup_{x \in \mathbb{R}^d} \left\{ \lambda_i x \cdot y - \frac{\lambda_i}{2}|x|^2 + g_i(x) \right\} \quad (2.11)$$

Hence, as a Legendre transform, the function $\frac{\lambda_i}{2}|\cdot|^2 - f_i$ is convex for $i \leq p-1$. Similarly, using that

$f(0) = 0$, we have:

$$g_i^n(x) \leq \frac{\lambda_i}{2}|x|^2 \quad \forall i \quad (2.12)$$

Since f^n is maximizing, there exists $C_1 \in \mathbb{R}$ such that:

$$\sum_{i=1}^p \int_{\mathbb{R}^d} g_i^n(x) dv_i(x) \geq C_1 \quad \forall n \quad (2.13)$$

Using (2.12) and (2.13), we have for all j :

$$\int_{\mathbb{R}^d} g_j^n(x) dv_j(x) \geq C_1 - \sum_{i \neq j} \int_{\mathbb{R}^d} g_i^n(x) dv_i(x) \quad (2.14)$$

Then, since $v_i \in X'$, it follows that the sequence $(\int_{\mathbb{R}^d} g_j^n(x) dv_j(x))_n$ is bounded. Then:

$$\begin{aligned} f_i^n(x) &\leq \frac{\lambda_i}{2}|x - y|^2 - g_i^n(x) \\ &\leq \frac{\lambda_i}{2}|y|^2 + \frac{\lambda_i}{2} \int_{\mathbb{R}^d} |x|^2 - 2x \cdot y dv_i(x) - \int_{\mathbb{R}^d} g_i^n(x) dv_i(x) \\ &\leq C_2(1 + |y|^2) \end{aligned}$$

where we used our bound on $(\int_{\mathbb{R}^d} g_j^n(x) dv_j(x))_n$ for the existence of C_2 . Using this upper bound as well as the fact that $\sum_1^n f_i = 0$, we have:

$$f_i^n(x) \geq -(p-2)C_2(1 + |y|^2)$$

This shows that $(f_i^n)_n$ is bounded in Y . Let (K_n) be an exhaustion of $\mathbb{R}^d$ by compact sets (for instance the closed balls of integer radius). Since a sequence of convex functions uniformly bounded on B_{2r} is equi-Lipschitz on B_r , the sequence $(\frac{\lambda_i}{2}|\cdot|^2 - f_i^n)_n$ is therefore equi-Lipschitz on every compact set. Moreover, for fixed x , $(\frac{\lambda_i}{2}|x|^2 - f_i^n(x))$ is bounded by the upper bounds obtained with C_2 . By the Arzelà-Ascoli theorem, for every compact set K , there exists a subsequence of $(\frac{\lambda_i}{2}|\cdot|^2 - f_i^n)_n$ converging for the Y norm on K . Removing the constant term, for every K , there therefore exists $f_i \in Y$ such that (f_i^n) converges in Y to f_i . We can use a diagonal extraction to obtain a subsequence converging on every K_n to the same limit f_i , and then (f_i^n) converges pointwise to f_i on $\mathbb{R}^d$. This allows us to write:

$$\begin{aligned} \limsup_n S_{\lambda_i} f_i^n(x) &= \limsup_n g_i^n(x) \leq \inf_{y \in \mathbb{R}^d} \limsup_n \left\{ \frac{\lambda_i}{2}|x - y|^2 - f_i^n(y) \right\} \\ &= \inf_{y \in \mathbb{R}^d} \left\{ \frac{\lambda_i}{2}|x - y|^2 - f_i(y) \right\} \\ &= S_{\lambda_i} f_i(x). \end{aligned}$$

Integrating, it only remains to justify that $\limsup_n \int S_{\lambda_i} f_i^n(x) dv_i(x) \leq \int \limsup_n S_{\lambda_i} f_i^n(x) dv_i(x)$, which is obtained by applying Fatou's lemma to $\frac{\lambda_i}{2}|\cdot|^2 - S_{\lambda_i} f_i^n$, which is nonnegative. $\blacksquare$

Remark 2.14. This proof highlights the interest of (P^*) and (P_0^*) . On the one hand, (P_0^*) is necessary because S_λ preserves X but not Y . On the other hand, the sequence (f^n) is bounded in Y but not in X . Hence (P_0^*) is necessary to prove the existence of a maximizer, which exists in (P^*) .

2.3 Uniqueness of the solution

The following theorem concludes this part on problems (P) and (P^*) :

Theorem 2.15: Uniqueness

If there exists an index i such that ν_i is absolutely continuous with respect to the Lebesgue measure, then (P) admits a unique solution ν given by:

$$\nu = \nabla \phi_i \# \nu_i$$

where $\lambda_i \phi_i(x) = \frac{\lambda_i |x|^2}{2} - S_{\lambda_i} f_i(x)$ and $(f_1, \dots, f_p)$ is a solution of (P^*) .

Proof. Let ν be a solution of (P) and, for $i \in \{1, \dots, p\}$, let ϕ_i be the functions such that $\lambda_i \phi_i(x) = \frac{\lambda_i |x|^2}{2} - S_{\lambda_i} f_i(x)$. The ϕ_i are convex and l.s.c. Let γ_i be the optimal transport plan from ν_i to ν . If we show that its support is included in $\partial \phi_i$, we will then have, using [theorem 1.3](#), $\gamma_i = (\text{id}, \nabla \phi_i) \# \nu_i$. In particular, $\nu = \nabla \phi_i \# \nu_i$, which determines ν uniquely. We do not prove the announced inclusion here, the proof being essentially computational and in the style of what was done above. ■

3. Link with the marginal formulation

The goal of this part is to introduce another optimization problem (Q) , which seems *a priori* different from the initial problem (P) , and to present several results showing that this problem is in fact closely related to (P) . More precisely, we will explain how, knowing a solution of the dual problem (Q^*) , one can deduce the barycenter, i.e. the solution of (P) .

For $x = (x_1, \dots, x_p) \in (\mathbb{R}^d)^p$, we set:

$$T(x) := \sum_{i=1}^p \lambda_i x_i \quad (3.1)$$

This is the Euclidean barycenter of the x_i associated with the weights λ_i . In other words:

$$\sum_{i=1}^p \lambda_i |x_i - T(x)|^2 = \inf_{y \in \mathbb{R}^d} \left\{ \sum_{i=1}^p \lambda_i |x_i - y|^2 \right\}$$

The optimal multi-marginal transport problem is the following:

$$\inf \left\{ \int_{(\mathbb{R}^d)^p} \left(\sum_{i=1}^p \frac{\lambda_i}{2} |x_i - T(x)|^2 \right) d\gamma(x_1, \dots, x_p) : \gamma \in \Pi(\nu_1, \dots, \nu_p) \right\}$$

Expanding the squared norm, we note that it is equivalent to the problem:

$$(Q) : \sup \left\{ \int_{(\mathbb{R}^d)^p} \left(\sum_{1 \leq i < j \leq p} \lambda_i \lambda_j x_i \cdot x_j \right) d\gamma(x_1, \dots, x_p) : \gamma \in \Pi(\nu_1, \dots, \nu_p) \right\} \quad (3.2)$$

The algorithms for solving (Q) are too complex. We therefore introduce the dual problem:

Proposition 3.1: Dual

The dual problem is the problem:

$$(Q^*) : \inf \left\{ \sum_{i=1}^p \int_{\mathbb{R}^d} g_i dv_i : \sum_{i=1}^p g_i(x_i) \geq \sum_{1 \leq i < j \leq p} \lambda_i \lambda_j x_i \cdot x_j, \forall x \in (\mathbb{R}^d)^p \right\} \quad (3.3)$$

The following theorem shows that knowing a solution of (Q^*) allows us to solve (Q) :

Theorem 3.2

If, for all i , v_i is absolutely continuous with respect to the Lebesgue measure, then there exists a unique $\gamma \in \Pi(v_1, \dots, v_p)$ that solves (Q) . Moreover, $\gamma = (T_1^1, \dots, T_p^1) \# v_1$, where $T_i^1 = \nabla u_i^* \circ \nabla u_1$ and

$$u_i(x) = \frac{\lambda_i}{2} |x|^2 + \frac{g_i(x)}{\lambda_i} \quad (3.4)$$

with $g_1, \dots, g_p$ convex functions that solve (Q^*) .

The following theorem shows that knowing a solution of (Q) allows us to solve (P) :

Theorem 3.3

If, for all i , v_i is absolutely continuous with respect to the Lebesgue measure, then the unique solution of (P) is given by $\nu = T \# \gamma$, with γ given by [theorem 3.2](#) and T the Euclidean barycenter given by [\(3.1\)](#).

Combining these last two theorems, we obtain:

Theorem 3.4

When the v_i are absolutely continuous with respect to the Lebesgue measure, the barycenter of the v_i associated with the weights λ_i is given by:

$$\nu = \left(\sum_{i=1}^p \lambda_i T_i^1 \right) \# v_1 \quad (3.5)$$

where T_i^1 is defined by [theorem 3.2](#).

Remark 3.5. This formula is analogous to the well-known case of the Euclidean barycenter, since $T_i^1 \# v_1 = v_i$.

Remark 3.6. Of course, since the roles of the v_j are symmetric, we deduce, for all j :

$$\nu = \left(\sum_{i=1}^p \lambda_i T_i^j \right) \# v_j \quad (3.6)$$

where T_i^j is defined by $T_i^j = \nabla u_i^* \circ \nabla u_j$.

4. Examples

Theorem 3.4 and theorem 2.15 seem to give formulas for computing the barycenters of the ν_i associated with the weights λ_i . But these formulas involve the solutions of problems (P^*) and (Q^*) , for which we have not presented any solution method. The goal of this part is thus to give explicit solutions in particular cases.

4.1 The case $d = 1$

We denote by F_i the cumulative distribution function of ν_i and we define its generalized inverse:

Definition 4.1

$$F_i^{(-1)}(t) := \inf \{ \alpha : F_i(\alpha) \geq t \} \quad (4.1)$$

Remark 4.2. Since $F_i : \mathbb{R} \rightarrow [0, 1]$ is nondecreasing, the generalized inverse is well defined for $t \in [0, 1]$, and if $t \in]0, 1[$, then it is finite.

Remark 4.3. From the definition, one can check that the generalized inverse is nondecreasing.

This allows us to make the barycenter explicit in the case $d = 1$:

Theorem 4.4

Suppose that the measures ν_i are all absolutely continuous with respect to the Lebesgue measure. The barycenter of the ν_i associated with the weights λ_i is given by

$$\nu = \left(\sum_{i=1}^p \lambda_i F_i^{(-1)} \circ F_1 \right) \# \nu_1 \quad (4.2)$$

To facilitate the proof of this result, we state three lemmas:

Lemma 4.5

Let F be a cumulative distribution function. For $x \in [0, 1]$ and $a \in \overline{\mathbb{R}}$:

$$F^{(-1)}(x) \leq a \Leftrightarrow x \leq F(a)$$

Proof. The reverse implication is an immediate consequence of the definition of the generalized inverse. For the direct implication, let $\alpha := F^{(-1)}(x)$. There therefore exists a sequence (α_n) of reals to the right of α satisfying $F(\alpha_n) \geq x$ with $\alpha_n \rightarrow \alpha$. By right-continuity of F , $F(\alpha) \geq x$. Moreover, by assumption, $\alpha \leq a$, so by monotonicity of F , $F(\alpha) \leq F(a)$. Hence $F(a) \geq x$. ■

Lemma 4.6

Let μ be a probability measure on $\mathbb{R}$ and u the measure associated with the uniform distribution on $[0, 1]$. We have:

$$F_\mu^{(-1)} \# u = \mu$$

Proof. $F_\mu^{(-1)} \# u(\cdot - \infty, t] = u(F_\mu^{(-1)^{-1}}(\cdot - \infty, t])$. But lemma 4.5 indicates that $x \in F_\mu^{(-1)^{-1}}(\cdot - \infty, t]$ if and only if $F_\mu^{(-1)}(x) \leq t$, i.e. $x \in] - \infty, F_\mu(t)]$. Hence $u(F_\mu^{(-1)^{-1}}(\cdot - \infty, t]) = u(\cdot - \infty, F_\mu(t)] = F_\mu(t)$. ■

Lemma 4.7

Let μ be a probability measure on $\mathbb{R}$ which is diffuse (i.e. without atoms) and u the measure associated with the uniform distribution on $[0, 1]$. Then:

1. $F_{\mu \#} \mu = u$
2. $\forall l, \{x : F_\mu(x) = l\}$ is μ -negligible

Proof. The second point is a consequence of the first. For the first, we note that since μ is diffuse, F is continuous, and since it is moreover nondecreasing, for $t \in]0, 1[$, $F_\mu^{-1}(\cdot - \infty, t]$ is an interval of the form $] - \infty, x_t[$, with $F_\mu(x_t) = t$. Thus, $F_{\mu \#} \mu(\cdot - \infty, t] = \mu(\cdot - \infty, x_t] = F_\mu(x_t) = t$. ■

We can now prove theorem 4.4:

Proof. First, we note that, using the uniqueness of the barycenter (theorem 2.15) and the characterization given by theorem 3.4, it suffices to show that $T_i^1 = F_i^{(-1)} \circ F_1$. For this, we note that T_i^1 is a nondecreasing map (as a composition of functions that are nondecreasing as gradients of convex functions). Moreover, this map sends v_1 to v_i . We will show that there is only one map satisfying these two conditions and that $F_i^{(-1)} \circ F_1$ works.

Admissibility: $F_i^{(-1)} \circ F_1$ is finite v_1 -a.e. Indeed, if $F_1(x) \in]0, 1[$, this is Remark 4.2, and lemma 4.7 shows that, up to a μ -null set, we can neglect the other case. We can therefore consider $F_i^{(-1)} \circ F_1 \# v_1$. Moreover, $F_i^{(-1)} \circ F_1$ is nondecreasing as a composition of nondecreasing functions. Finally, using successively lemma 4.7 and lemma 4.6, we have:

$$F_i^{(-1)} \circ F_1 \# v_1 = F_i^{(-1)} \# (F_1 \# v_1) = F_i^{(-1)} \# (u) = v_i$$

Uniqueness: Let T be a nondecreasing map such that $T \# v_1 = v_i$. By monotonicity of T : $] - \infty, x] \subset T^{-1}(\cdot - \infty, T(x))$. Hence, using $T \# v_1 = v_i$, we get $F_1(x) \leq F_i(T(x))$. Using the definition of $F^{(-1)}$, we therefore have $F_i^{(-1)} \circ F_1(x) \leq T(x)$. Let x be such that the previous inequality is strict. Setting $\varepsilon_0 := T(x) - F_i^{(-1)} \circ F_1(x)$, we have, for all $\varepsilon \in]0, \varepsilon_0[$: $T(x) - \varepsilon \geq T(x) - \varepsilon_0 = F_i^{(-1)} \circ F_1(x)$. Using lemma 4.5: $\forall \varepsilon \in]0, \varepsilon_0[, F_i(T(x) - \varepsilon) \geq F_1(x)$. But on the other hand, by monotonicity of T : $T^{-1}(\cdot - \infty, T(x) - \varepsilon) \subset] - \infty, x[$, which implies $F_i(T(x) - \varepsilon) \leq F_1(x)$. Thus, for $\varepsilon \in]0, \varepsilon_0[$, $F_i(T(x) - \varepsilon)$ is constant and equal to $F_1(x)$. But since these intervals are at most countable, the values $F_1(x)$ such that x satisfies $F_i^{(-1)} \circ F_1(x) < T(x)$ are at most countable. Such an x is therefore contained in $\bigcup_i \{x, F_1(x) = l_i\}$, which has measure zero by lemma 4.7. Hence $F_i^{(-1)} \circ F_1(x) = T(x)$ v_1 -a.e. But in the expression $T \# v_1 = v_i$, T can be defined up to a v_1 -negligible set, which therefore shows uniqueness. ■

4.2 The case $p = 2$

In the case where there are only two measures, the barycenter takes the following form:

Theorem 4.8

Let ν_0 and ν_1 be measures of $X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$ that are absolutely continuous with respect to the Lebesgue measure, and $t \in]0, 1[$. The barycenter of (ν_0, ν_1) associated with the weights $(1 - t, t)$ is given by ν_t , where:

$$\nu_t := ((1 - t)\text{id} + t\nabla\phi) \# \nu_0 = (t\text{id} + (1 - t)\nabla\phi^*) \# \nu_1$$

with $\nabla\phi$ the Brenier map from ν_0 to ν_1 .

This theorem gives us the shape of the geodesics in the space $X' \cap \mathcal{M}_+^1(\mathbb{R}^d)$. Moreover, it is important since it explains how to interpolate between two measures of this space in a way that is optimal in a certain sense.

The proof of this theorem essentially reduces to the proof of the two lemmas:

Lemma 4.9

$$W_2(\nu_0, \nu_t) = tW_2(\nu_0, \nu_1) \quad (4.3)$$

$$W_2(\nu_t, \nu_1) = (1 - t)W_2(\nu_0, \nu_1) \quad (4.4)$$

Proof. Let us set $T_t := (1 - t)\text{id} + t\nabla\phi$. We need to show that $(\text{id}, T_t) \# \nu_0$ is an optimal transport plan from ν_0 to ν_t , since we will then have

$$W_2^2(\nu_0, \nu_t) = \int_{\mathbb{R}^d} |x - T_t(x)|^2 d\nu_0(x) = t^2 \int_{\mathbb{R}^d} |x - \nabla\phi(x)|^2 d\nu_0(x) = t^2 W_2^2(\nu_0, \nu_1)$$

We will use [theorem 1.3](#). Given the definition of the measure under consideration, the support of $(\text{id}, T_t) \# \nu_0$ is $\{(x, T_t(x)) : x \in \mathbb{R}^d\}$. But since $T_t = \nabla\Phi$, with $\nabla\Phi = \frac{1-t}{2}|\cdot|^2 + t\phi$ (which is convex and l.s.c.), the support of $(\text{id}, T_t) \# \nu_0$ is nothing other than $\partial\Phi$. [Theorem 1.3](#) concludes the first equality. For the second, we do the same proof by considering the map $t\text{id} + (1 - t)\nabla\phi^*$. ■

Lemma 4.10

Let $c \in \mathbb{R}_+$. The function $(a, b) \mapsto \frac{1-t}{2}a^2 + \frac{t}{2}b^2$ under the constraint $a + b \geq c$ admits a minimum $\frac{t(1-t)}{2}c^2$.

Proof. This result can be proved using, for example, Lagrangian duality. We define the Lagrangian $L(a, b, \lambda) := \frac{1-t}{2}a^2 + \frac{t}{2}b^2 + \lambda c - \lambda(a + b)$. Its gradient (with respect to a, b) vanishes when $(1 - t)a = \lambda = tb$. We then maximize with respect to λ the quantity $\frac{-\lambda^2}{2(1-t)} + \frac{\lambda^2}{2t} + \lambda c$. The maximum is attained for $\lambda = t(1 - t)c$. Finally, we compute $L(tc, (1 - t)c, t(1 - t)c) = \frac{t(1-t)c^2}{2}$. ■

Let us now prove [theorem 4.8](#).

Proof. Let ν be admissible. Set $a := W_2(\nu_0, \nu)$, $b := W_2(\nu, \nu_1)$ and $c := W_2(\nu_0, \nu_1)$. With these notations, the functional $J(\nu)$ equals $\frac{1-t}{2}a^2 + \frac{t}{2}b^2$. The triangle inequality imposes $a + b \geq c$. But using [lemma 4.9](#), we get $J(\nu_t) = \frac{t(1-t)}{2}c^2$. [Lemma 4.10](#) yields the first equality of the theorem. For

the second, we note that $\nabla\phi^*$ is the Brenier map from ν_1 to ν_0 . It therefore suffices to apply the first result proved in the theorem to $\nu'_0 = \nu_1$ and $\nu'_1 = \nu_0$. ■

4.3 The Gaussian case

In this part, we assume that for $i = 1, \dots, p$, $\nu_i \sim \mathcal{N}(0, S_i)$, where $S_i \in \mathcal{S}_n^{++}(\mathbb{R})$. We want to find the barycenter of these Gaussians, i.e. to solve (P). It is given by the following theorem:

Theorem 4.11

There exists a symmetric positive definite matrix S satisfying:

$$\sum_{i=1}^p \lambda_i \left(S^{1/2} S_i S^{1/2} \right)^{1/2} = S \quad (4.5)$$

Problem (P) admits a unique solution ν given by $\nu \sim \mathcal{N}(0, S)$.

Remark 4.12. *This theorem gives two important pieces of information. The barycenter of several centered Gaussians is a centered Gaussian. It therefore suffices to determine its covariance matrix to obtain an explicit expression of the barycenter. This amounts to solving the matrix equation (4.5).*

To prove this theorem, we recall Brouwer's fixed point theorem:

Theorem 4.13: Brouwer fixed point

Let $f : K \rightarrow K$ be a continuous function, where K is a nonempty compact convex subset of a Euclidean space; then f has a fixed point.

We also take for granted the following characterization, which can be proved from [theorem 2.15](#):

Theorem 4.14

Suppose that all the measures in (P) are absolutely continuous with respect to the Lebesgue measure, and that, for a measure ν admissible for (P) and absolutely continuous with respect to the Lebesgue measure, $\nabla\phi_i$ is the Brenier map sending ν_i to ν . If the equality:

$$\sum_{i=1}^p \lambda_i \nabla\phi_i^* = \text{id} \quad (4.6)$$

holds ν -a.e., then ν is a solution of (P).

Let us now prove [theorem 4.11](#).

Proof. Existence of S : Let α_i and β_i be respectively the smallest and the largest eigenvalue of S_i . Let α and β be two reals such that:

$$\beta \geq \left(\sum_{i=1}^p \lambda_i \sqrt{\beta_i} \right)^2 \geq \sum_{i=1}^p (\lambda_i \sqrt{\alpha_i})^2 \geq \alpha$$

Let $K_{\alpha, \beta}$ be the set of matrices $S \in \mathcal{S}_n^{++}$ such that $\beta I \geq S \geq \alpha I$, where we say that $A \geq B$ if

$A - B \in \mathcal{S}_n^+$. $K_{\alpha,\beta}$ is convex. Moreover, it is closed and bounded, hence compact. Let us define the following function F :

$$F(S) := \sum_{i=1}^p \lambda_i \left(S^{1/2} S_i S^{1/2} \right)^{1/2}$$

We then have:

$$\beta I \geq \sum_{i=1}^p \lambda_i \sqrt{\beta \beta_i} I \geq F(S) \geq \sum_{i=1}^p \lambda_i \sqrt{\alpha \alpha_i} I \geq \alpha I, \quad \forall S \in K_{\alpha,\beta}$$

Indeed, to obtain the first inequality, we need to show that for all $x \in \mathbb{R}^d$, we have $\langle \beta I x, x \rangle \geq \langle \sum_{i=1}^p \lambda_i \sqrt{\beta \beta_i} I x, x \rangle$, which amounts to showing that $\beta \geq \sum_{i=1}^p \lambda_i \sqrt{\beta \beta_i}$, which is true by definition of β . Similarly, we need to show that $\forall x, \sum_{i=1}^p \lambda_i \sqrt{\beta \beta_i} \|x\|^2 \geq \langle F(S)x, x \rangle$. But we know from the spectral theorem that, for a real symmetric matrix Σ , we have $(\forall x, \lambda \|x\|^2 \geq \langle \Sigma x, x \rangle)$ iff $\lambda_{\max} \leq \lambda$, where $\lambda_{\max}$ denotes the largest eigenvalue of Σ . The matrix $(S^{1/2} S_i S^{1/2})$ is real symmetric positive. But: $\langle S^{1/2} S_i S^{1/2} x, x \rangle = \langle S_i (S^{1/2} x), S^{1/2} x \rangle \leq \beta_i \|S^{1/2} x\|^2 \leq \beta_i \beta \|x\|^2$. Hence its largest eigenvalue is smaller than $\beta_i \beta$. Hence the largest eigenvalue of $(S^{1/2} S_i S^{1/2})^{1/2}$ is smaller than $\sqrt{\beta_i \beta}$, which concludes. The other two inequalities are analogous.

We deduce that F is a map from $K_{\alpha,\beta}$ into itself, which is moreover continuous given its definition. [Theorem 4.13](#) then gives the existence of a solution to equation (4.5).

Let us show that $\nu \sim \mathcal{N}(0, S)$ solves (P): Let $\nu \sim \mathcal{N}(0, S)$ with S a solution of (4.5). By [theorem 1.7](#), T_i is the Brenier map sending ν to ν_i , where:

$$T_i := S_i^{1/2} \left(S_i^{1/2} S S_i^{1/2} \right)^{-1/2} S_i^{1/2} \quad (4.7)$$

By [theorem 4.14](#) as well as [theorem 1.6](#), it suffices to show that $\sum_{i=1}^p \lambda_i T_i = I$. For this, let us denote $K_i := S_i^{1/2}$ and $K := S^{1/2}$. We use the equality:

$$(K K_i^2 K)^{1/2} = K K_i (K_i K^2 K_i)^{-1/2} K_i K$$

which is proved by squaring both sides and invoking the uniqueness of the square root in $\mathcal{S}_n^{++}$. The equality $F(S) = S$ then reads:

$$\sum_{i=1}^p \lambda_i K K_i (K_i K^2 K_i)^{-1/2} K_i K = K^2$$

Finally, we use the invertibility of K :

$$\sum_{i=1}^p \lambda_i K_i (K_i K^2 K_i)^{-1/2} K_i = \sum_{i=1}^p \lambda_i T_i = I$$

Uniqueness: We already know that (P) admits a unique solution ([theorem 2.15](#)). By the previous step, if S is a solution of (4.5), then $\mathcal{N}(0, S)$ is a solution of (P). Hence the solution of (4.5) is unique. ■

5. Numerical approach

Here we seek to compute barycenters numerically using the above.

5.1 The case $d = 1$

Lemma 4.6 allows us to reformulate equation 4.2 as: $\nu = \left(\sum_{i=1}^p \lambda_i F_i^{(-1)} \right) \# u$. Lemma 4.6 then shows that $F_\nu^{(-1)} = \sum_{i=1}^p \lambda_i F_i^{(-1)}$. If we take strictly increasing F_i , the functions under consideration are bijective and thus $F_\nu^{-1} = \sum_{i=1}^p \lambda_i F_i^{-1}$. If moreover the F_i are sufficiently regular (C^1), then ν has density $f_\nu = \frac{1}{Q' \circ Q^{-1}}$, where we denote by $Q := \sum_{i=1}^p \lambda_i F_i^{-1}$ the quantile function of the barycenter ν (a change of variables suffices to prove this result). Numerically, we have access to the quantiles of the usual distributions, hence to $Q(t)$. We can estimate Q' by finite differences and plot the graph $\{(Q(t), 1/Q'(t)), t \in \mathbb{R}\}$, which is indeed the graph of f_ν since Q is bijective.

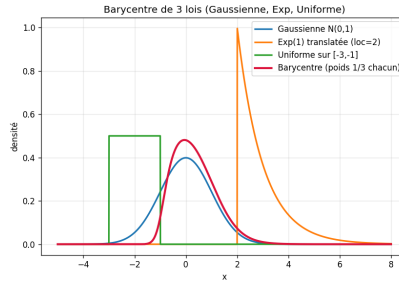
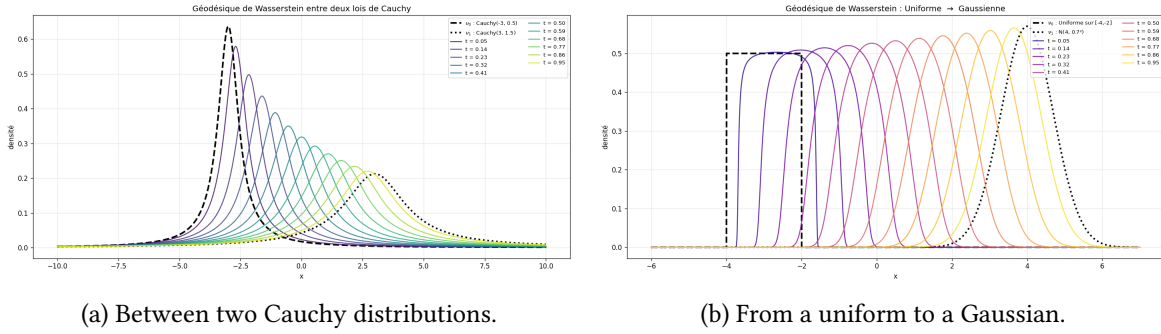


Figure 5.1: Barycenter of three distributions on $\mathbb{R}$ (Gaussian, exponential, uniform, $\lambda_i = 1/3$).



(a) Between two Cauchy distributions.

(b) From a uniform to a Gaussian.

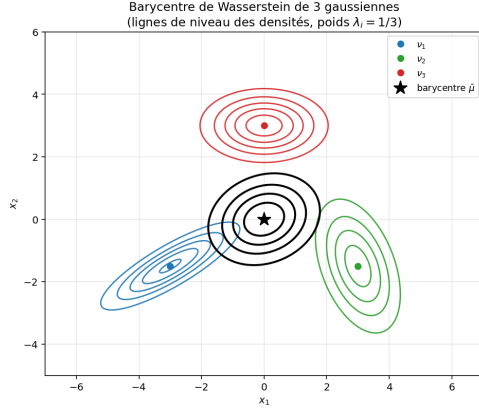
Figure 5.2: Wasserstein geodesics between two measures on $\mathbb{R}$, computed using the procedure described above, without using theorem 4.8. Cauchy distributions do not have a second moment, but formula 4.2 seems to remain valid here.

5.2 The Gaussian case

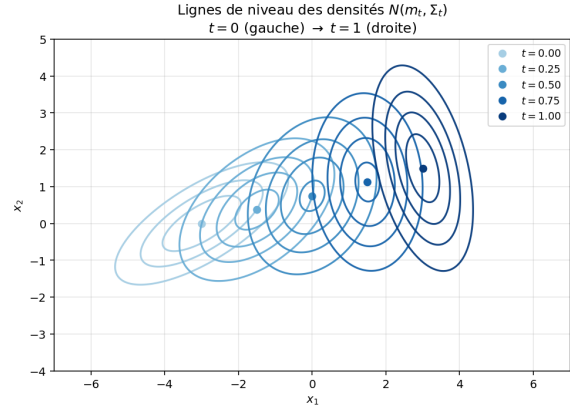
In the Gaussian case, one can show that the solution of equation 4.5 can be approximated with the iterative formula:

$$S_{n+1} = S_n^{-1/2} \left(\sum_{j=1}^p \lambda_j (S_n^{1/2} \Sigma_j S_n^{1/2})^{1/2} \right)^2 S_n^{-1/2}$$

One can refer to the article [3] for more details.



(a) Barycenter of three Gaussians on $\mathbb{R}^2$ with weights $\lambda_i = 1/3$.



(b) Wasserstein geodesic between two Gaussians on $\mathbb{R}^2$.

Figure 5.3: Illustration of the Gaussian case (theorem 4.11) in dimension 2.

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